Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Acceleration

Meaning in my words How would you explain acceleration to a friend?	**How can you tell?** A cart follows a bend at steady speed. Why is it accelerating?
Example or sketch Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits?	**Non-example and why** A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Velocity

Meaning in my words How would you explain velocity to a friend?	**How can you tell?** Right is positive. What does a velocity of −2 m/s tell you?
Example or sketch How would you show a cart going east at 3 m/s?	**Non-example and why** A speedometer reads 3 m/s. What would you still need for velocity?

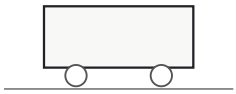
Speed trend

Meaning in my words How would you explain speed trend to a friend?	**How can you tell?** A cart moves left but accelerates right. What happens to its speed before it stops?
Example or sketch Velocities one second apart are −6, −4 and −2 m/s. Describe the trend.	**Non-example and why** A student says −2 to −6 m/s means the cart slowed. Which numbers should they compare?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw an object with two opposing forces. Write the signed sum under the net arrow.



My marks mean

Moving left. Pushed right.

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish motion, net force, and acceleration.

CHOOSE ONE JOB TOGETHER

CONTROLS ENGINEER ☐

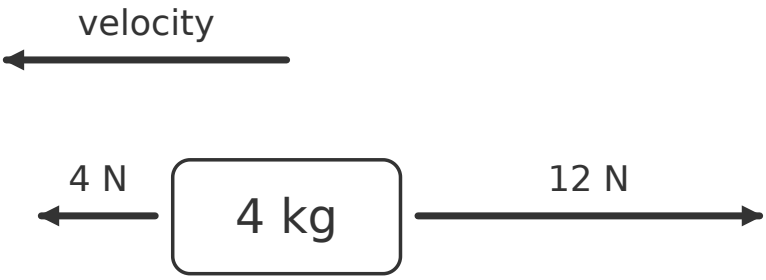
Draw the net-force and acceleration arrows. Tell the operator what happens to speed while the body still moves left.

REPLAY ANALYST ☐

Label separate arrows for motion and acceleration. Explain to viewers how they can point opposite ways.

Point to a clue. Keep your first marks; add the update using the color key.

A tracking model moves left. Its **4 kg** body feels **12 N** right and **4 N** left. The status display has separate motion and force arrows.



01 OUR FIRST CALL

What happens to speed at first? Include the net force and acceleration in your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the updated display. What does a force arrow tell you, and what does it not tell you?

Switch speakers. What additional information would let you predict when the body stops and reverses?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks